

Alpha-Szegő Rank-One Dynamics: Exact Cascades and Isospectral Blindness

HCS-C386 theorem and reproducibility package

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Abstract

We close the full rank-one alpha-Szegő threshold with one physical clock, an explicit radial orbit, and its exact high-frequency consequence. A conserved energy defect separates relatively compact trajectories from the positive-perturbation cascade level. On that level a directly derived second-order equation gives a two-sided hyperbolic-secant-squared profile, including its turning point. For every Sobolev exponent above one half, the logarithmic norm-growth rate is the positive radial exponent multiplied by the distance of that Sobolev exponent from one half. The critical norm remains constant while momentum leaves every fixed Fourier window. The native shifted-Hankel square has only one nonzero eigenvalue and a degree-one Fredholm determinant. At fixed positive perturbation a bounded pure mode and a cascading solution share this determinant, so auxiliary isospectrality does not determine even the recurrence type. Exact rational reconstruction and independent high-precision calculations audit the source formulas. The classical threshold is credited to Xu; no arithmetic target, generic Hardy-data classification, or Hilbert-Pólya conclusion is claimed.

中文摘要

本文在同一物理时间下闭合塞戈方程的完整秩一阈值、显式径向轨道与高频级联。

守恒能量缺陷区分紧轨道与正扰动级联，直接二阶方程保持转折点唯一性。

级联径向量具有双向双曲正割平方公式，高阶索伯列夫范数的指数增长率精确确定。

临界范数保持不变，而守恒动量逐渐离开每个固定傅里叶窗口。

原生移位汉克尔算子的平方只有一个非零特征值，其行列式不能区分同参数下的有界轨道与级联轨道。

有理重建、独立高精度积分和尾界核查仅作为回归证据；经典结果明确归属，不提出算术目标或希尔伯特波利亚结论。

Keywords: alpha-Szegő; rank-one dynamics; energy threshold; turning point; Sobolev cascade; isospectrality

关键词：塞戈方程；秩一动力学；能量阈值；转折点；索伯列夫级联；等谱性

1 One invariant nonlinear source

A finite-rank invariant manifold need not be a finite Fourier approximation. Here it contains infinitely many modes and an exact nonlinear evolution. The cubic Szegő equation and its Hankel structure originate in Gérard-Grellier [1]; Xu [2] established the rank-one threshold for its linear perturbation. Our contribution within this repository is a complete convention-preserving proof and reproducibility package. In particular, we retain the rank hypothesis, the turning point, and the distinction between an auxiliary determinant and physical evolution. We do not claim literature priority for the classical threshold.

Use normalized circle measure and let Π retain nonnegative Fourier modes. For every real α fix

$$iu_t = \Pi(|u|^2 u) + \alpha(u \mid 1), \quad u(z) = b + \frac{cz}{1-pz}, \quad c \neq 0, \quad |p| < 1. \quad (1)$$

The phase space in (1) is $\mathcal{L}(1)$; constants will be treated separately. The physical time is t , without rescaling. Set

$$d = 1 - |p|^2, \quad \rho = |c|, \quad B = |b|^2, \quad Q = B + \frac{\rho^2}{d}, \quad M = \frac{\rho^2}{d^2} > 0. \quad (2)$$

Energy and its defect are

$$E_\alpha = \frac{1}{4} \|u\|_4^4 + \frac{\alpha}{2} |b|^2, \quad \delta = E_\alpha - \frac{Q^2}{4} - \frac{\alpha Q}{2}. \quad (3)$$

We always take the positive square root of M . On a trajectory, $\rho = \sqrt{M}d$ and $B = Q - Md$.

Theorem 1 (Exact all-mode reduction and global existence). *Every initial datum in $\mathcal{L}(1)$ has a unique solution for all real time. Its coefficients satisfy*

$$\begin{aligned} i\dot{b} &= (B + 2\rho^2/d + \alpha)b + \rho^2 c\bar{p}/d^2, \\ i\dot{c} &= (2B + \rho^2/d^2)c + 2b\rho^2 p/d, \\ i\dot{p} &= c\bar{b} + \rho^2 p/d. \end{aligned} \quad (4)$$

The quantities Q, M, E_α are conserved. These statements hold at $b = 0$ and $p = 0$ and involve no Fourier cutoff.

Proof. The geometric Fourier coefficients are $u_0 = b$ and $u_n = cp^{n-1}$ for $n \geq 1$. Their absolute convergence on strict subdisks justifies convolution in $\Pi(|u|^2 u)$. Compare the coefficients of $1, z/(1-pz), z^2/(1-pz)^2$; the derivative of the last coordinate is $c\dot{p}$, not merely $\dot{p}$. The result is (4). A direct quartic sum gives

$$4E_\alpha = B^2 + \frac{4B\rho^2}{d} + \frac{\rho^4(1+|p|^2)}{d^3} + \frac{4\rho^2\Re(bp\bar{c})}{d^2} + 2\alpha B. \quad (5)$$

Differentiating (2) and (5) along (4) cancels every term, giving the three conservation laws.

To justify all-time continuation, put $X = bp\bar{c}$. Direct differentiation gives

$$\dot{d} = 2\Im X, \quad \dot{\rho} = 2\sqrt{M}\Im X, \quad |\dot{\rho}/\rho| \leq 2\sqrt{QM}. \quad (6)$$

Hence $\rho(t) \geq \rho(0)e^{-2\sqrt{QM}|t|}$ on every finite time interval. The conserved quantities bound $|b| \leq \sqrt{Q}, \rho \leq \sqrt{M}$, and $d = \rho/\sqrt{M}$ away from zero on that interval. The real six-dimensional vector field is locally Lipschitz on $d > 0$, so its continuation theorem excludes finite-time exit. The resulting geometric series solves (1) in every H^s . This argument proves the named invariant-manifold theorem, not global well-posedness for arbitrary Hardy data. $\square$

2 Compactness and the exact exceptional faces

The defect is useful because it vanishes precisely where a compactness estimate could fail.

Lemma 1 (Factored defect and a quantitative bound). *For every trajectory in $\mathcal{L}(1)$,*

$$\delta = \frac{Md}{2} \left(\left| b + \frac{c\bar{p}}{d} \right|^2 - \alpha \right). \quad (7)$$

If $\delta \neq 0$, then for every real t ,

$$d(t) \geq \frac{2|\delta|}{M((\sqrt{Q} + \sqrt{M})^2 + |\alpha|)} > 0. \quad (8)$$

The orbit is relatively compact in $\mathcal{L}(1)$ and bounded in every H^s .

Proof. Subtract the two terms in (3) from (5); the remaining quadratic expression is (7). Since $|b + c\bar{p}/d| \leq \sqrt{Q} + \sqrt{M}$, taking absolute values proves (8). Thus $|p|$ lies in a strict subdisk and $|c| = \sqrt{M}d$ stays positive, while b, c remain bounded. These inequalities give compact closure within the coefficient manifold. Every Sobolev norm is then bounded by a convergent geometric series. $\square$

For $\alpha < 0$, (7) has strictly positive defect; thus all rank-one orbits are relatively compact. On a compact set of initial data, the positive continuous defect has a positive minimum, making the bound uniform for that set. This does not assert uniformity across data approaching a positive-alpha zero-defect hypersurface.

For $\alpha = 0$, nonzero defect is already settled. Zero defect forces $b = -c\bar{p}/d$. Writing $A = c/d$ yields

$$Q = M = |A|^2, \quad u(z) = A \frac{z - \bar{p}}{1 - pz}, \quad u(t, z) = e^{-iQt} u(0, z). \quad (9)$$

The quotient in (9) has modulus one on the circle. Substitution in (4) gives $\dot{p} = 0, \dot{b} = -iQb$ and $\dot{c} = -iQc$, proving the formula even at $p = 0$. Therefore every zero-alpha rank-one orbit is also relatively compact.

Rank zero cannot inherit the threshold. If $c = 0$, the parameter p is redundant and the complete solution is

$$u(t, z) = b(0)e^{-i(|b(0)|^2 + \alpha)t}. \quad (10)$$

Every constant satisfies $\delta = 0$ but has constant Sobolev norm. The zero function is stationary. A nonzero constant is stationary exactly when $|b|^2 + \alpha = 0$; otherwise its least temporal period is $2\pi/||b|^2 + \alpha|$. These facts explain why the rank-one hypothesis cannot be dropped from a positive-alpha growth criterion.

3 Turning-point-safe cascade

Assume $\alpha > 0$ and $\delta = 0$. Define the discriminant

$$\kappa^2 = 4QM - (\alpha - Q - M)^2. \quad (11)$$

The next result strengthens the usable form of the classical threshold: it provides the complete radial trajectory with a valid turning-point proof.

Theorem 2 (Complete positive-alpha dichotomy). *For initial data in $\mathcal{L}(1)$ and $\alpha > 0$, the following are equivalent: the trajectory is not relatively compact in $\mathcal{L}(1)$; and $\delta = 0$. On that level $\kappa > 0$, and, with*

$$d_* = \frac{\kappa^2}{4\alpha M}, \quad t_* = \frac{2}{\kappa} \operatorname{artanh} \left(\frac{\dot{d}(0)}{\kappa d(0)} \right), \quad (12)$$

the exact radial orbit for every real t is

$$d(t) = d_* \operatorname{sech}^2 \left(\frac{\kappa(t - t_*)}{2} \right), \quad \rho(t) = \sqrt{M} d(t). \quad (13)$$

Here $0 < d_* \leq 1$, and the formula includes $p = 0$ at the turn.

Proof. Equation (7) gives $2\Re X = d(\alpha - Q - M) + 2Md^2$. Together with $|X|^2 = (Q - Md)Md^2(1 - d)$ and (6), it implies

$$\dot{d}^2 = d^2(\kappa^2 - 4\alpha Md). \quad (14)$$

Since $d > 0$, the right side forces $\kappa^2 \geq 4\alpha Md > 0$. Moreover,

$$\kappa^2 - 4\alpha M = -(Q - M - \alpha)^2 \leq 0, \quad (15)$$

so $d_* \leq 1$.

A squared velocity identity alone permits spurious sticking at a turn. To remove that ambiguity, differentiate the complex ODE itself. Put $T = \alpha - Q - M$. Then

$$\begin{aligned} \dot{X} &= i(FX + G), & F &= -T - 4Md, \\ G &= -M^2 d^2(1 - d) - (Q - Md)Md^2 + 2(Q - Md)Md(1 - d). \end{aligned} \quad (16)$$

All coefficients F, G are real. Since $\dot{d} = 2\Im X$, substitute $\Re X = dT/2 + Md^2$ in $\ddot{d} = 2(F\Re X + G)$ to obtain the polynomial equation

$$\ddot{d} = \kappa^2 d - 6\alpha Md^2. \quad (17)$$

This derivation never divides by $\dot{d}$ or by p . At $d = d_*$ it gives $\ddot{d} = -\kappa^2 d_*/2 < 0$, excluding a stationary turning point.

By (14), $v_0 = \dot{d}(0)/(\kappa d(0))$ lies strictly between -1 and 1 , so (12) is finite. The function (13) solves (17). At time zero its logarithmic derivative is $\kappa \tanh(\kappa t_*/2) = \kappa v_0$. Also $\operatorname{sech}^2(\kappa t_*/2) = 1 - v_0^2 = d(0)/d_*$, so both initial data agree. Uniqueness for the polynomial second-order initial-value problem proves (13) for every real t , including the turn.

The radial variable tends to zero at both infinite-time ends, so this orbit does not have compact closure within $\mathcal{L}(1)$. Every nonzero-defect orbit is compact by (8). This proves both directions of the dichotomy. $\square$

The all-time clause matters. No finite-time singularity is created: $d(t)$ remains positive for every finite t . The expression $\kappa^2/(4\alpha M)$ can equal one, but that is the harmless event $p = 0$, not the forbidden pole boundary $|p| = 1$.

The source threshold appears in Xu [2, Theorem 3.1]. We do not rely on a globally positive lower bound for $|\dot{d}/d|$: that derivative is zero at $t = t_*$. Equations (16)–(17) and uniqueness provide the complete argument instead.

4 All-mode Sobolev growth and momentum escape

Define the exact norm convention

$$\|u\|_{H^s}^2 = \sum_{n \geq 0} (1 + n)^{2s} |u_n|^2. \quad (18)$$

The radial theorem has an all-frequency consequence, not merely a growing finite-dimensional diagnostic.

Theorem 3 (Exact growth exponent). *On the positive-alpha cascade level, for every $s > 1/2$,*

$$\|u(t)\|_{H^s}^2 \sim M\Gamma(2s+1)d(t)^{1-2s}, \quad t \rightarrow \pm\infty, \quad (19)$$

$$\lim_{t \rightarrow \pm\infty} \frac{\log \|u(t)\|_{H^s}}{|t|} = (s - \frac{1}{2})\kappa. \quad (20)$$

At the critical exponent, $\|u(t)\|_{H^{1/2}}^2 = Q + M$ exactly.

Proof. For $d \downarrow 0$, the geometric sum satisfies

$$d^{2s+1} \sum_{n \geq 1} (n+1)^{2s} (1-d)^{n-1} \rightarrow \int_0^\infty x^{2s} e^{-x} dx = \Gamma(2s+1). \quad (21)$$

To justify this Riemann-sum limit, use $x = dn$. On a bounded interval separated from zero, the summand converges uniformly to $x^{2s} e^{-x}$. A multiple of x^{2s} controls the interval near zero, and an exponential majorant controls the tail. These bounds allow the two omitted intervals to be made arbitrarily small, uniformly for sufficiently small d .

Since $|c|^2 = Md^2$, multiplying (21) gives (19); the bounded constant coefficient does not contribute to the diverging leading term. Equation (13) yields

$$d(t) \sim \frac{\kappa^2}{\alpha M} e^{-\kappa|t-t_*|}, \quad (22)$$

and then (20) follows. At $s = 1/2$, direct geometric summation in (18) gives $Q + M$, proving the critical assertion. $\square$

The conserved momentum does not remain in a fixed frequency window. Indeed,

$$\|u(t) - b(t)\|_2^2 = Md(t) \rightarrow 0, \quad \sum_{n \geq 1} n|u_n(t)|^2 = M. \quad (23)$$

For each fixed N , the sum over $1 \leq n \leq N$ tends to zero because $|c(t)|^2 \rightarrow 0$. Thus the source preserves momentum while its location moves beyond every fixed Fourier cutoff. There is no finite-time norm singularity or loss of a conserved quantity.

For the explicit data $u_0(z) = \sqrt{\alpha} + z$, one has $Q = 1 + \alpha$, $M = 1$, $\kappa = 2\sqrt{\alpha}$, $t_* = 0$. Hence $d(t) = \text{sech}^2(\sqrt{\alpha}t)$ and the rate is $(2s-1)\sqrt{\alpha}$, agreeing with the classical example [2, Theorem 3.2].

5 The same determinant sees opposite dynamics

The relevant Hankel operator is antilinear. Fix the inner product linear in its first variable and define $H_u f = \Pi(u\bar{f})$, $K_u = S^* H_u$. The matrix of K_u is $(u_{j+k+1}) = (cp^{j+k})$. With $v = (1, p, p^2, \dots)$, direct multiplication gives

$$K_u f = c\langle v, f \rangle v, \quad K_u^2 f = \frac{|c|^2}{d} \langle f, v \rangle v. \quad (24)$$

Since $\|v\|^2 = 1/d$, K_u^2 is complex-linear, positive, rank one, and its sole nonzero eigenvalue is M .

Proposition 1 (Auxiliary isospectral blindness). *For all complex w and integers $r \geq 1$,*

$$\det(I - wK_u^2) = 1 - wM, \quad \text{tr}(K_u^{2r}) = M^r. \quad (25)$$

At each fixed $\alpha > 0$, this determinant is shared by a bounded pure-mode trajectory and a two-sided cascading trajectory.

Proof. The first assertions follow from the explicit rank-one linear operator in (24) and conservation of M . For $u_0 = z$, the source solution is $u(t) = e^{-it}z$, with $M = 1$. For $u_0 = \sqrt{\alpha} + z$, the preceding theorem gives a cascade, also with $M = 1$. Both therefore have determinant $1 - w$. Even the recurrence distinction is absent from that determinant. Allowing α to vary further keeps $1 - w$ fixed while the growth exponent $2\sqrt{\alpha}$ varies continuously. $\square$

Equation (25) is a native auxiliary invariant. It is not a determinant of the nonlinear time evolution, and it is not a primitive-orbit product. We do not assign a complex Fredholm determinant directly to the antilinear K_u . A zero-free entire factor cannot change the fact that $1 - wM$ has only one zero; this polynomial is not a full target divisor.

The source has a concrete reversal on Hardy functions:

$$(Cu)(z) = \overline{u(\bar{z})}. \quad (26)$$

It conjugates Fourier coefficients and preserves the Hardy space. The real convolution coefficients in (1) show that $Cu(-t)$ solves the same alpha equation. Pointwise conjugation alone would reverse Fourier support and is not the stated Hardy-space reversal. This nonlinear Hamiltonian symmetry is not a construction of a linear quantum Hamiltonian.

6 Evidence boundary and conclusion

Source face	Defect	Established behavior
Rank one, $\alpha < 0$	positive	relatively compact
Rank one, $\alpha = 0$	nonzero	relatively compact
Rank one, $\alpha = 0$	zero	inner-function phase orbit
Rank one, $\alpha > 0$	nonzero	relatively compact
Rank one, $\alpha > 0$	zero	exact two-sided cascade
Rank zero	always zero	constant-norm phase orbit

The regime table records a rank distinction, not a fitted classifier. All complex phases are retained in the coefficient ODE. The all-time and all-frequency conclusions are proved analytically; finite computations test independent reconstructions of their formulas.

The executable ledger contains 144 generic rational rows, 72 threshold rows, 12 zero-alpha inner rows, 24 constant rows and three paired controls. A Fraction-pair producer is checked by an independent SymPy reconstruction using a factored energy expression and the differentiated p equation. Thirteen symbolic identities pass. Independent 100-digit complex ODE integration tests 12 endpoints in both time directions; its largest radial error is below 3×10^{-52} against the exact profile. Twelve direct Fourier sums are checked against generating functions with geometric tail bounds. These are numerical regression results, not interval certificates. Two isolated producer trees reproduce the JSON bytes; 31 hostile cases, including three actual release-write YAML attacks, are rejected, and three smoke tests pass.

The strict assessment is

$$(A0_FAIL, A1_FAIL, A2_FAIL, A3_FAIL, A4_FORMAL_HINT).$$

Source integrability and an exact determinant do not supply an arithmetic origin, an arithmetic primitive census, or a target analytic structure.

The claim remains the rank-one source theorem. No arbitrary Hardy-data or higher-rank threshold, complete generic primitive census, target arithmetic local data, target Euler factors, root number, automorphy, target divisor, target functional equation, target-zero matching, or Hilbert-Pólya operator is claimed. Route B remains disabled. The firewall is NO_BAD_EULER_OR_ROOT_NUMBER.

Literature ownership and reproducibility. The classical equation and threshold belong to the cited sources. Our package makes the turning-point argument, rank-zero failure, physical clock and auxiliary-determinant limitation auditable together. No publication acceptance, worldwide novelty, external-human review or cross-provider validation is asserted. AI-assisted local drafting and independent collaborating-agent review are disclosed. The source proof, locked evaluation, independent scripts and three substantive PDF revisions accompany the article.

Conclusion. Round two: Sobolev cascade and determinant blindness. The explicit radial law fixes all high-Sobolev exponents, while a native isospectral determinant misses even the bounded-versus-cascading distinction.

References

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